Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Big Data Engineering, Distributed Data Processing, Streaming & Ingestion, Data Lakehouse Architecture, Data Modeling & Warehousing, Dimensional Modeling, Performance Tuning & Scalability, Autonomous Anomaly Detection, Generative AI & Prompt Orchestration, Predictive Machine Learning, MLOps, Automated Data Quality & Governance, Time-Series Forecasting

Programming Languages & Tools: Apache Spark, PySpark, Apache Flink, Kafka, Trino, Presto, Apache Iceberg, Snowflake, Databricks, Azure (ADF, Synapse, Azure SQL), AWS (S3, Redshift), Power BI, Tableau, GitHub Copilot, KIRO IDE, Collibra, Airflow

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data & AI Systems Engineer — **May 2022 – Present**

- **Architected and scaled distributed data pipelines** using Apache Spark, PySpark, and Azure/AWS cloud lakehouses to process hundreds of terabytes with 99.9% availability SLAs, aligning with large-scale big data engineering requirements.
- **Engineered autonomous agentic workflows and pipelines** utilizing KIRO IDE, prompt orchestration, and GitHub Copilot to automate schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50%.
- **Deployed real-time anomaly detection frameworks** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, ensuring high data platform integrity for downstream streaming and analytical systems.
- **Built and optimized high-throughput analytical workloads** incorporating dimensional data modeling (Star/Snowflake schemas, SCD Type 1/2) and performance-tuned SQL jobs to support low-latency querying and executive decision intelligence.
- **Integrated automated testing and evaluation frameworks** for conversational AI and distributed microservices platforms, cutting manual QA turnaround by 50% and accelerating robust production release cycles.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Developed predictive machine learning models** using XGBoost, Random Forests, and Scikit-Learn to forecast customer behavior, delivering actionable signals that improved operational service reliability by 20%.
- **Architected advanced time-series forecasting models** using SARIMA to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing response times by 20%.
- **Leveraged AI-assisted development tools** such as Copilot and Codex for accelerated debugging, query optimization, and rapid prototyping of complex data processing jobs.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed enterprise Key Operating Metric (KOM) dashboards** and semantic models in Power BI (DAX) and Tableau, translating complex transactional telemetry into executive corporate strategy.
- **Engineered automated data quality and governance firewalls** alongside Collibra data lineage integration, successfully reducing operational reconciliation turnaround by over 80%.
- **Optimized cloud data warehouse architectures** across Snowflake, Azure Synapse, and Redshift to streamline reporting pipelines and enhance analytics reliability.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.